

Introduction to Real Analysis – Homework 2

Lebesgue outer measure

Notation. m^* denotes Lebesgue outer measure. Write $E \triangle F = (E \setminus F) \cup (F \setminus E)$.

1. **Translation and scaling.** For $E \subset \mathbb{R}^d$, $v \in \mathbb{R}^d$ and $\lambda \neq 0$, write $v + \lambda E = \{v + \lambda x : x \in E\}$. Prove directly from box coverings that

$$m^*(v + \lambda E) = |\lambda|^d m^*(E).$$

Your proof should also cover $m^*(E) = \infty$.

2. **An estimate for changing a set.** Let $E, F \subset \mathbb{R}^d$ have finite outer measure. Prove

$$|m^*(E) - m^*(F)| \leq m^*(E \triangle F).$$

3. **Points belonging to infinitely many sets.** Suppose $E_n \subset \mathbb{R}^d$ and $\sum_{n=1}^{\infty} m^*(E_n) < \infty$. Prove that

$$m^*\left(\bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n\right) = 0.$$

Hint: bound the outer measure by each tail of the series.

4. **Challenge: a dense open set of small outer measure.** Given $0 < \varepsilon < 1$, construct an open dense set $U \subset \mathbb{R}$ with $m^*(U) < \varepsilon$. Then prove

$$m^*([0, 1] \setminus U) > 1 - \varepsilon.$$

Hint: enumerate the rationals and cover them by short open intervals.